VLSI Task 2: ADC Interface

Verilog RTL and Testbench — Xilinx Vivado

Add the RTL as a Design Source named **adc_interface.v** and the testbench as a Simulation Source named **tb_adc_interface.v**.

1. Verilog RTL Code

```
module adc_interface (
    input clk,
    input rst_n,
    input [7:0] adc_data,
    input sample_valid,
    output reg [7:0] sensor_data,
    output reg data_ready
);

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            sensor_data <= 8'd0;
            data_ready  <= 1'b0;
        end
        else begin
            data_ready <= 1'b0;
            if (sample_valid) begin
                sensor_data <= adc_data;
                data_ready  <= 1'b1;
            end
        end
    end

endmodule
```

2. Verilog Testbench Code

```
 `timescale 1ns / 1ps

 module tb_adc_interface;
     reg clk = 0;
     reg rst_n = 0;
     reg [7:0] adc_data = 0;
     reg sample_valid = 0;

     wire [7:0] sensor_data;
     wire data_ready;

     adc_interface uut (
         .clk(clk),
         .rst_n(rst_n),
         .adc_data(adc_data),
         .sample_valid(sample_valid),
         .sensor_data(sensor_data),
         .data_ready(data_ready)
     );

     always #5 clk = ~clk;

     initial begin
         #20;
         rst_n = 1;

         // Sample 1: decimal 100
         @(negedge clk);
         adc_data = 8'd100;
         sample_valid = 1;
         @(negedge clk);
         sample_valid = 0;

         // Sample 2: decimal 200
         @(negedge clk);
         adc_data = 8'd200;
         sample_valid = 1;
         @(negedge clk);
         sample_valid = 0;

         // Sample 3: decimal 50
         @(negedge clk);
         adc_data = 8'd50;
         sample_valid = 1;
         @(negedge clk);
         sample_valid = 0;

         #20;
         $finish;
     end
 endmodule
```

Expected behavior

After reset is released, the testbench sends decimal samples 100, 200 and 50. When sample_valid is HIGH at a rising clock edge, sensor_data captures adc_data and data_ready is HIGH for one clock cycle. When sample_valid is LOW, sensor_data retains its last captured value.

Note

This is a digital sample-capture interface for simulation practice. It does not implement the serial protocol of a physical ADC chip.